

NanoSafe Analyzer — Research & Cytotoxicity Report

Evaluation of the Biocompatibility and Cytotoxicity of ZnO Nanoparticles

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1. EXPERIMENT OVERVIEW

- Experiment / Sample Name: Experiment test 1
- Cell Line Evaluated: HeLa
- Target Medical Application: General Biomedical Research
- Lead Researcher: Researcher
- Nanoparticle Type: ZnO
- Exposure Duration: N/A
- ISO 10993-5 Compliance: FAIL — Cytotoxic
- Data Source: Direct Entry

2. STUDY PARTICIPANT & BIOLOGICAL SAMPLE TRACEABILITY

- Biological Sample Traceability: Standard in-vitro reference cell culture line.
- Participant Consent: In-vitro cell line — anonymized laboratory experiment.

3. AVERAGED CYTOTOXICITY & BIOMARKER METRICS

- Mean Cell Viability: 71.7%
- Reactive Oxygen Species (ROS): 2.84
- Apoptosis Inducement: 6.14%
- Mean ZnO Concentration: 37.0 µg/mL
- LDH Membrane Leakage: 9.459999999999999%

4. MACHINE LEARNING CYTOTOXICITY ASSESSMENT

- Predicted Toxicity Score: 11.51 / 100
- Estimated IC50 (Half-Max Inhibitory): 58.01 µg/mL
- Risk Level Classification: Low
- Biocompatible Safe Window: 0.0 - 15.7 µg/mL

5. SCIENTIFIC INTERPRETATION & RECOMMENDATION

Local Scikit-Learn Ensemble Model evaluated ZnO cytotoxicity on HeLa cells under a 24.0h exposure duration for 'Wound Dressing & Topical Antimicrobial'. Measured parameters: Viability=71.7%, ROS=2.84, LDH=9.459999999999999%, Apoptosis=6.14%. The offline ML model computed a Toxicity Score of 11.51 out of 100 (Low Risk Zone, ISO 10993-5 status: CONDITIONAL — Low-Dose Monitoring Required). Predicted IC50: 44.83 µg/mL, Safe Biomedical Dosage Ceiling: 0.0 - 15.7 µg/mL. Clinical Rationale: Evaluates dermal